CO1 – ASSESSMENT 2

ANALYTICAL PROBLEM SOLVING

Analyse the Mathematics of a Diabetic Retinopathy Screening Model (RetinaCare)

Total Marks: 50 | Bloom's Level: K4 – Analyse | Nature: Individual (In-class)

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All numerical answers are reported to four decimal places. Information-theoretic logarithms use base 2.

TASK 1 – Break Down the Audit Problem

Data	Concept	Model stage
D1	SVD and rank-1 approximation	Representation / compression
D2	Hessian, convexity, gradient descent	Loss / optimisation
D3	Entropy, cross-entropy, KL divergence	Probability evaluation
D4	Bayes' rule, predictive values	Clinical decision
D5	Bias–variance decomposition	Prediction-error evaluation
D6	Confusion-matrix metrics	Classification evaluation

Dependencies: D1 determines representation quality; D2 governs optimiser convergence; D3 assesses probability quality; D4 combines sensitivity, specificity and prevalence; D5 quantifies prediction error; D6 assesses empirical classification mistakes. D4 and D6 describe different operating characteristics and should not be conflated.

Assumptions: H is constant and symmetric; probabilities sum to one; the four predictions are equally weighted; and sensitivity/specificity in the design challenge remain fixed as stated.

TASK 2 – Examine Feature-Matrix Compressibility

$A = [[3, 0], [4, 5]]$. $A^TA = [[25, 20], [20, 25]]$.

$\det(A^TA - \lambda I) = (25 - \lambda)^2 - 400 = (\lambda - 45)(\lambda - 5) = 0$. Hence $\lambda_1 = 45$, $\lambda_2 = 5$ and $\sigma_1 = \sqrt{45} = 6.7082$, $\sigma_2 = \sqrt{5} = 2.2361$; $\sigma_1/\sigma_2 = 3.0000$.

Normalised right singular vectors: $v_1 = (1, 1)/\sqrt{2}$, $v_2 = (1, -1)/\sqrt{2}$. From $u_i = Av_i/\sigma_i$: $u_1 = (1, 3)/\sqrt{10}$ and $u_2 = (3, -1)/\sqrt{10}$.

Thus $A = U\Sigma V^T$, with $U = [[0.3162, 0.9487], [0.9487, -0.3162]]$, $\Sigma = \text{diag}(6.7082, 2.2361)$, and $V^T = [[0.7071, 0.7071], [0.7071, -0.7071]]$.

$A_1 = \sigma_1 u_1 v_1^T = [[1.5, 1.5], [4.5, 4.5]]$. Residual $A - A_1 = [[1.5, -1.5], [-0.5, 0.5]]$.

$\|A - A_1\|F = \sqrt{(1.5^2 + 1.5^2 + 0.5^2 + 0.5^2)} = \sqrt{5} = 2.2361$. Energy retained $= 45/(45 + 5) \times 100 = 90.0000\%$.

Conclusion: Rank-1 compression retains 90.0000% of squared singular-value energy, but the lost 10.0000% may include clinically relevant signals. Validate diagnostic performance before adopting compression.

TASK 3 – Analyse Loss-Surface Curvature

$H = [[4, 1], [1, 4]]$, $b = (5, 5)^T$. $\det(H - \lambda I) = (4 - \lambda)^2 - 1 = 0$, so $\lambda_1 = 5$ and $\lambda_2 = 3$.

Corresponding unit eigenvectors are $(1, 1)/\sqrt{2}$ (high curvature) and $(1, -1)/\sqrt{2}$ (low curvature). Both eigenvalues are positive, so H is positive definite and J is strictly convex with a unique minimum.

Condition number $\kappa_2(H) = 5/3 = 1.6667$: the surface is well conditioned. $H^{-1} = (1/15)[[4, -1], [-1, 4]]$, so $w^* = H^{-1}b = (1, 1)^T$.

Verification: $\nabla J(w^*) = Hw^* - b = (5, 5)^T - (5, 5)^T = (0, 0)^T$. Minimum loss $J(w^*) = -5.0000$.

TASK 4 – Information-Theoretic Comparison

$p = [0.5, 0.25, 0.125, 0.125]$, $q_1 = [0.25, 0.25, 0.25, 0.25]$, $q_2 = [0.4, 0.3, 0.2, 0.1]$.

$H(p) = -\Sigma p_i \log_2 p_i = 0.5 + 0.5 + 0.375 + 0.375 = 1.7500$ bits. $H(q_1) = -4(0.25\log_2 0.25) = 2.0000$ bits. $H(p,q_1) = -\Sigma p_i \log_2 q_{1i} = 2.0000$ bits.

Class	$\mathbf{p_i \log_2(p_i/q_{1i})}$	$\mathbf{p_i \log_2(p_i/q_{2i})}$
No DR	0.5000	0.1610
Mild	0.0000	−0.0658
Moderate	−0.1250	−0.0848
Severe	−0.1250	0.0402
Total	0.2500	0.0506

$KL(p \| q_1) = 0.2500$ bits; $KL(p \| q_2) = 0.0506$ bits. Verify $H(p,q_1) = H(p) + KL(p \| q_1) = 1.7500 + 0.2500 = 2.0000$ bits.

The No DR class contributes the largest positive term in each comparison. Individual KL terms can be negative although total KL cannot. q_2 is closer to p; zero KL means identical distributions on all classes with positive p.

TASK 5 – Bayesian Screening Decision

Prevalence $P(R) = 0.10$, sensitivity $P(+|R) = 0.90$, specificity $P(-|\neg R) = 0.95$. Probability-tree branches: R (0.10) → + (0.90), − (0.10); ¬R (0.90) → + (0.05), − (0.95). Joint probabilities: TP 0.0900, FN 0.0100, FP 0.0450, TN 0.8550.

$P(R|+) = (0.90 \times 0.10)/[(0.90 \times 0.10) + (0.05 \times 0.90)] = 0.09/0.135 = 0.6667 = 66.6667\%$.

$P(R|-) = (0.10 \times 0.10)/[(0.10 \times 0.10) + (0.95 \times 0.90)] = 0.01/0.865 = 0.0116 = 1.1561\%$.

PPV is below 90% sensitivity because it also depends on prevalence and false positives. A positive screen warrants ophthalmologist assessment, not automatic treatment.

TASK 6 – Three Gradient-Descent Iterations

$\nabla J(w) = Hw - b$; $w_{t+1} = w_t - 0.1\nabla J(w_t)$, starting from $w_0 = (0, 0)$.

Step 1: $g_0 = (-5, -5)$; $w_1 = (0.5, 0.5)$; $J(w_1) = \frac{1}{2}(2.5) - 5 = -3.7500$.

Step 2: $g_1 = (-2.5, -2.5)$; $w_2 = (0.75, 0.75)$; $J(w_2) = \frac{1}{2}(5.625) - 7.5 = -4.6875$.

Step 3: $g_2 = (-1.25, -1.25)$; $w_3 = (0.875, 0.875)$; $J(w_3) = \frac{1}{2}(7.65625) - 8.75 = -4.9219$.

Step	w	J(w)	$\|w - w^*\|_2$
0	(0, 0)	0.0000	1.4142
1	(0.5, 0.5)	−3.7500	0.7071
2	(0.75, 0.75)	−4.6875	0.3536
3	(0.875, 0.875)	−4.9219	0.1768

Directional error factors are $1 - \eta\lambda_1 = 0.5000$ and $1 - \eta\lambda_2 = 0.7000$. The initial error lies along the $\lambda = 5$ eigenvector, so its distance halves each step. The loss approaches −5.0000.

TASK 7 – Momentum Versus Plain Gradient Descent

With $v_0 = 0$, $\beta = 0.9$ and $\eta = 0.1$: $v_{t+1} = \beta v_t - \eta \nabla J(w_t)$; $w_{t+1} = w_t + v_{t+1}$.

Step 1: $v_1 = (0.5, 0.5)$, $w_1 = (0.5, 0.5)$. Step 2: $v_2 = 0.9(0.5, 0.5) - 0.1(-2.5, -2.5) = (0.7, 0.7)$; $w_2 = (1.2, 1.2)$.

Step	Plain GD weights	Momentum weights
1	(0.5, 0.5)	(0.5, 0.5)
2	(0.75, 0.75)	(1.2, 1.2)

Momentum overshoots $w^* = (1, 1)$ because accumulated velocity persists. For plain GD, $|1 - \eta\lambda|_{max} < 1$ requires $0 < \eta < 2/5 = 0.4000$.

At $\eta = 0.5$: $w_1 = (2.5, 2.5)$, $g_1 = (7.5, 7.5)$, $w_2 = (-1.25, -1.25)$. Distances increase from 1.4142 initially to 2.1213 then 3.1820: the method diverges. This bound applies to plain GD; momentum has a different stability condition.

TASK 8 – Bias–Variance Decomposition

True score $y = 10$, irreducible noise $\sigma^2 = 1$. Use population variance (divide by 4) for the four equally weighted predictions. Expected error = Bias² + Variance + σ^2.

Model P = [8, 9, 8, 9]. Mean = 8.5000; Bias² = $(8.5 - 10)^2 = 2.2500$; variance = $[4(0.5^2)]/4 = 0.2500$; total = 2.25 + 0.25 + 1 = 3.5000.

Model Q = [12, 7, 11, 8]. Mean = 9.5000; Bias² = $(9.5 - 10)^2 = 0.2500$; variance = $(2.5^2 + 2.5^2 + 1.5^2 + 1.5^2)/4 = 4.2500$; total = 0.25 + 4.25 + 1 = 5.5000.

Component	Model P	Model Q
Bias²	2.2500	0.2500
Variance	0.2500	4.2500
Noise	1.0000	1.0000
Expected error	3.5000	5.5000

Model P is bias-dominated; Q is variance-dominated. P is preferable by expected squared error. The noise term is additional irreducible error, so the observed four-prediction mean squared errors themselves are 2.5000 and 4.5000.

TASK 9 – Confusion-Matrix Error Profile

Actual / Predicted	Referable	Non-referable

Referable	TP = 40	FN = 10
Non-referable	FP = 5	TN = 45

Accuracy = (40 + 45)/100 = 85.0000%. Error rate = (10 + 5)/100 = 15.0000%. Precision = 40/(40 + 5) = 88.8889%. Recall = 40/(40 + 10) = 80.0000%. Specificity = 45/(45 + 5) = 90.0000%. F1 = 2(40)/[2(40) + 5 + 10] = 84.2105%.

False negatives (10) exceed false positives (5), risking delayed diagnosis and preventable vision loss. Accuracy alone hides the distribution and relative clinical costs of errors.

TASK 10 – Overall Audit Findings

Compression: D1 rank-1 SVD retains 90.0000% energy, but clinically relevant information could be lost. Loss/optimisation: positive eigenvalues (5, 3), $\kappa = 1.6667$ and the convergence trace support the quadratic optimiser at $\eta = 0.1$. Probability quality: KL falls from 0.2500 to 0.0506 bits. Error: Model P has lower expected error (3.5000 vs 5.5000). Clinical concerns: D6 sensitivity is only 80.0000% with 10 missed referable cases. D4's 90%/95% are separate assumed operating characteristics.

Audit judgement: the mathematical design is promising, but compression and decision thresholds require clinical validation, especially to reduce missed referable cases.

TASK 11 – Design Challenge: Low Prevalence

A. Impact Analysis

Prevalence-dependent: PPV, P(R|−), NPV, expected counts per 1,000 and total error cost. Prevalence-independent under the scenario assumptions: sensitivity 90%, specificity 95%, D1 SVD, D2 Hessian and optimisation. Posterior probabilities change because Bayes' rule includes the prior prevalence.

B. Re-examination

At prevalence 0.02: P(R|+) = (0.90 × 0.02)/[(0.90 × 0.02) + (0.05 × 0.98)] = 0.018/0.067 = 0.2687 = 26.8657%.

P(R|−) = (0.10 × 0.02)/[(0.10 × 0.02) + (0.95 × 0.98)] = 0.002/0.933 = 0.0021 = 0.2144%.

Expected per 1,000	**10% prevalence**	**2% prevalence**
Actual referable	100	20
True positives	90	18
False negatives	10	2
False positives	45	49
True negatives	855	931

At 2% prevalence, FN = 20 × 0.10 = 2 and FP = 980 × 0.05 = 49. The absolute number of missed cases falls, but the proportion of positive results that are true cases falls sharply.

C. Policy Options and Cost

Require PPV ≥ 0.5: 0.5 = 0.018/[0.018 + 0.98(1 − Sp)]. Therefore 0.018 = 0.98(1 − Sp), giving Sp ≥ 1 − 0.018/0.98 = 0.98163265 = 98.1633%, provided sensitivity stays 90%.

Let false-negative cost = 10 and false-positive cost = 1. Cost per 1,000 = 10FN + FP. Baseline at 10%: 10(10) + 45 = 145 units; baseline at 2%: 10(2) + 49 = 69 units.

Option at 2% prevalence	Sensitivity	Specificity	Expected cost / 1,000
Retain baseline	90%	95%	69
Higher specificity (hypothetical)	90%	98.1633%	38
Perfect sensitivity (hypothetical)	100%	95%	49

Higher-specificity target: FP = 980 × (1 − 0.98163265) = 18; FN = 2; cost = 18 + 20 = 38. Perfect-sensitivity target: FP = 49, FN = 0; cost = 49. These hypothetical combinations are not established feasible threshold points.

Raising a threshold generally increases specificity but decreases sensitivity; lowering it generally increases sensitivity but increases false referrals. With a 10:1 cost ratio, a missed case is especially harmful. A proposed threshold must be evaluated using its actual paired sensitivity and specificity; the mathematically possible 38-unit scenario is not a guaranteed outcome.

D. Evidence-Based Conclusion

Decision: MODIFY the screening policy, with clinical validation before use. At 2% prevalence PPV is only 26.8657%, so a positive screen should prompt confirmatory ophthalmic examination rather than immediate treatment. Baseline error cost is 69 units per 1,000 (49 unnecessary referrals and 2 missed cases). A hypothetical specificity increase could lower this to 38 units only if sensitivity stays at 90%. Because missed cases cost ten times more, the operating threshold should be selected from measured sensitivity–specificity/cost curves, with follow-up for poor-quality images and symptomatic screen-negative patients. Rural constraints include specialist capacity, travel costs, image quality and reliable follow-up.

FINAL CONCLUSION

RetinaCare has a well-conditioned convex objective, stable gradient descent at $\eta = 0.1$ and potentially useful 90% energy-retaining feature compression. Nevertheless, the D6 false-negative burden and low-prevalence PPV require careful clinical validation. Modify the referral workflow and verify operating thresholds before deployment.